Exploring Vibration Simulated Braille for Accessible Mobile Messaging

Author: Swapnil Puri
Advisor: Dr. Bilal Shebaro
04/17/2026

What is Braille?

- Braille is a tactile system where patterns of raised and flat dots in a six-dot cell represent characters and are read by touch.
- Braille remains a fundamental and broadly utilized literacy tool that provides visually impaired individuals with essential access to written information.
- Braille literacy has been shown to be closely associated with individuals' self-esteem, sense of competence, and overall well-being.



The Problem

- Information is increasingly accessed through electronic devices, and many types of content are now available only in digital form.
- These devices lack integrated hardware for tactile braille output.
- As a result, visually impaired users must rely on workarounds to access information non-visually.



Existing “Solutions”

Refreshable Braille Display

- A standard 40-cell display can cost users between $3,500 and $15,000.
- Users have also reported concerns regarding portability and operational noise.

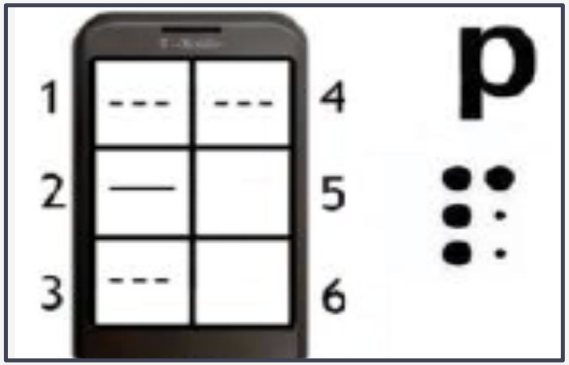




Audio-Based Technologies

- Not accessible for deafblind individuals.
- Not suitable in many social situations.
- Undermines the value of tactile literacy.

The New Solutions

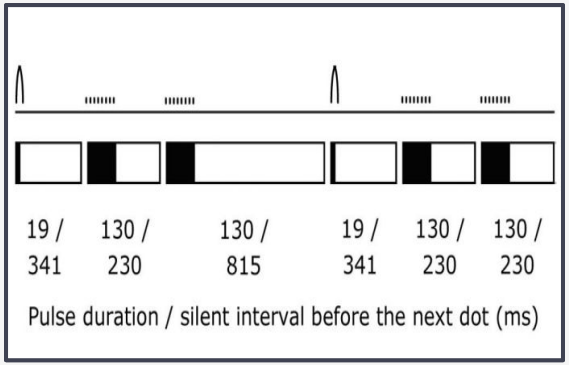


V-Braille

- V-Braille uses six on-screen touch regions that vibrate or stay still to represent the raised and flat dots of a braille cell (Jayant et al., 2010).
- Nine users achieved 90% letter-recognition accuracy with V-Braille.

Braille Rhythm

- Users feel a sequence of strong and weak vibrations as the system moves through the braille cell from top-left to bottom-left, then top-right to bottom-right.
- Resulted in greater accuracy and faster reading performance.





Issues with the New Solutions

- Designed primarily for experimental testing, not real-world use.
- Limited to simple tasks like single-letter recognition.
- Hampered by user experience issues such as accidental keypresses and uncomfortable interaction methods.



Research Gap

- No studies have developed a vibration-braille system that:
  - Functions as a full, usable mobile application.
  - Enables continuous access to written content.
  - Offers practical value that visually impaired users could rely on.

How can vibration-simulated braille be integrated into a mobile communication platform to enhance accessibility for visually impaired users?



The Solution

- Vibcom is an iOS-based, multimodal communication app designed for full accessibility.
- Users can compose messages using swipe-based Braille input, character drawing, or traditional typing.
- Messages can be received through haptic feedback, audio output, or visual text display.
- The system is built using SwiftUI, with Firebase supporting authentication, data storage, and real-time communication.

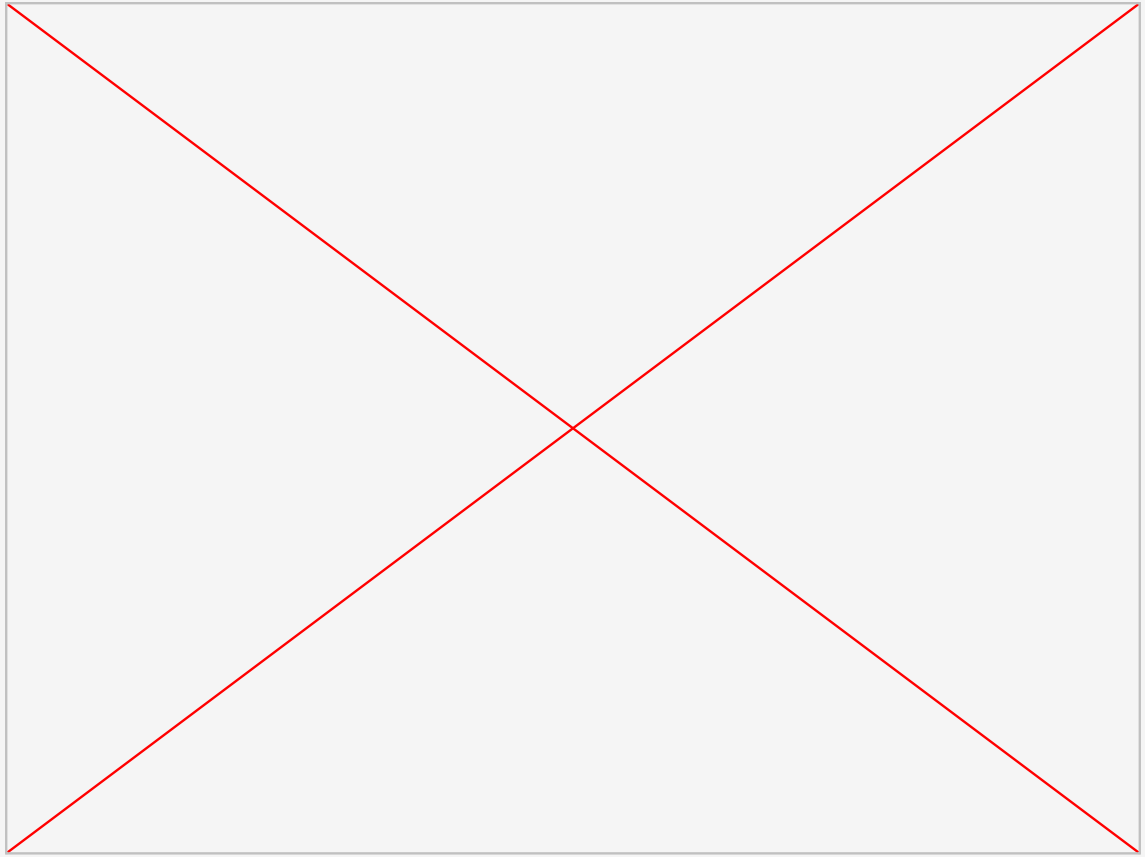


The Solution

- Vibconnect is an iOS app that enables users to easily discover others and exchange contact information for messaging in Vibcom.
- Users can find nearby individuals or share phone numbers directly when both devices are in close proximity.
- The app uses gesture-based interactions instead of physical buttons to ensure a more accessible experience.



Demonstration



Future Work



User Testing

Conduct usability studies with visually impaired users to validate and refine the system



More Accessibility Options

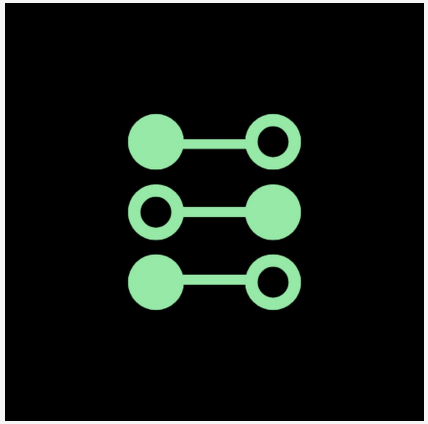
Allow customization of speech speed, audio settings, and vibration intensity/patterns



Enhanced Feature Set

Introduce group messaging and additional user customization features





Conclusion

- Vibcom and Vibconnect offer a more accessible approach to mobile messaging through multimodal, gesture-based input and vibration feedback for users of all visual abilities.

- They also highlight how mainstream platforms can adopt similar features to improve accessibility and reach a wider audience.

Thank You.
Questions?

References

American Foundation for the Blind. (2024). What is braille? https://www.afb.org/blindness-and-low-vision/braille/what-braille

Dolphin, S., Downing, M., Cirrincione, M., Samuta, A., Leite, K., Noble, K., & Walsh, B. (2024). Information accessibility in the form of Braille. IEEE Open Journal of Engineering in Medicine and Biology, 5, 205–209. https://doi.org/10.1109/OJEMB.2024.3364065

Jayant, C., Acuario, C., Johnson, W., Hollier, J., & Ladner, R. (2010). V-Braille: Haptic Braille perception using a touch-screen and vibration on mobile phones. In Proceedings of the 12th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '10) (pp. 295–296). Association for Computing Machinery. https://doi.org/10.1145/1878803.1878878

Martinez, J., & Kamei-Hannan, C. (2025). Preferences and opinions of braille users on the development of a multiline braille display. Journal of Blindness Innovation and Research, 15(1). https://nfb.org/images/nfb/publications/jbir/jbir25/jbir150101.html.

Rantala, J., et al. (2009). Methods for presenting Braille characters on a mobile device with a touchscreen and tactile feedback. IEEE Transactions on Haptics, 2(1), 28–39. https://doi.org/10.1109/TOH.2009.3

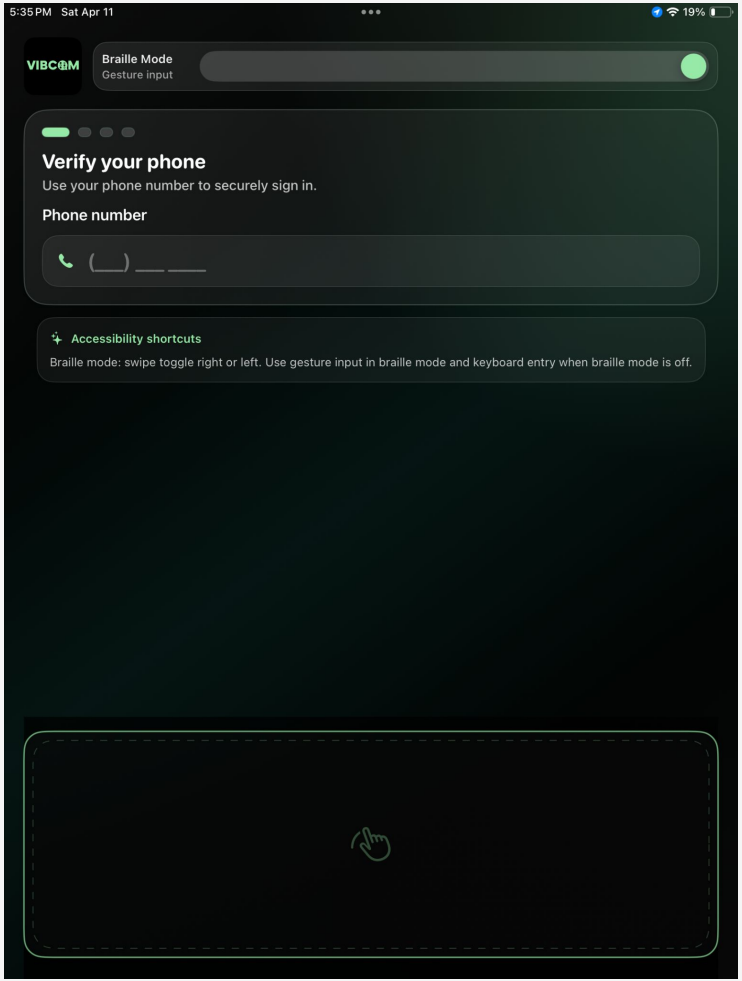
Silverman, A. M., & Bell, E. C. (2018). The association between braille reading history and well-being for blind adults. Journal of Blindness Innovation and Research, 8(1). https://nfb.org/images/nfb/publications/jbir/jbir18/jbir080103.html

Limitations

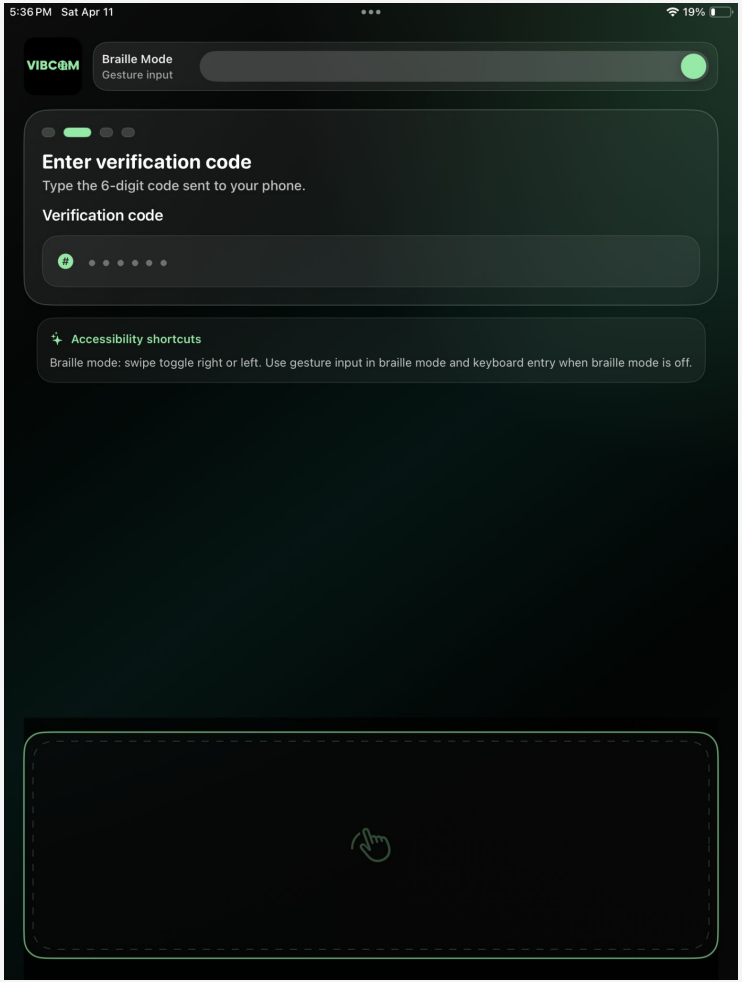
- No user testing conducted with visually impaired participants
- Lack of formal evaluation with the target user population
- Project focused primarily on system design and implementation



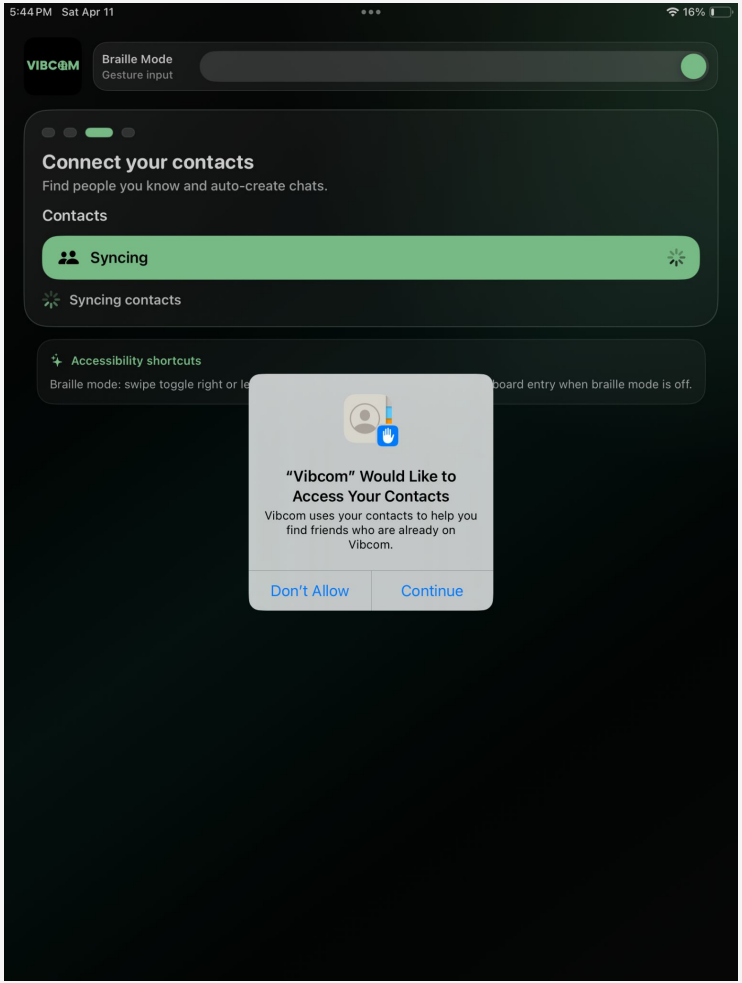
Vibcom Phone Verification



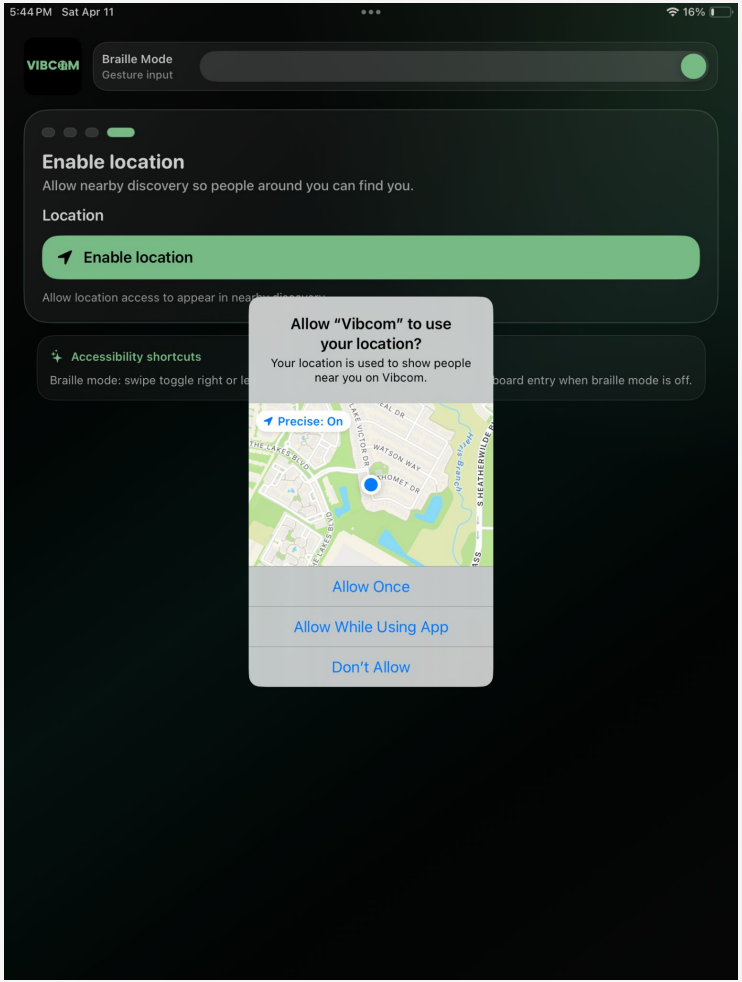
Vibcom Verification Code



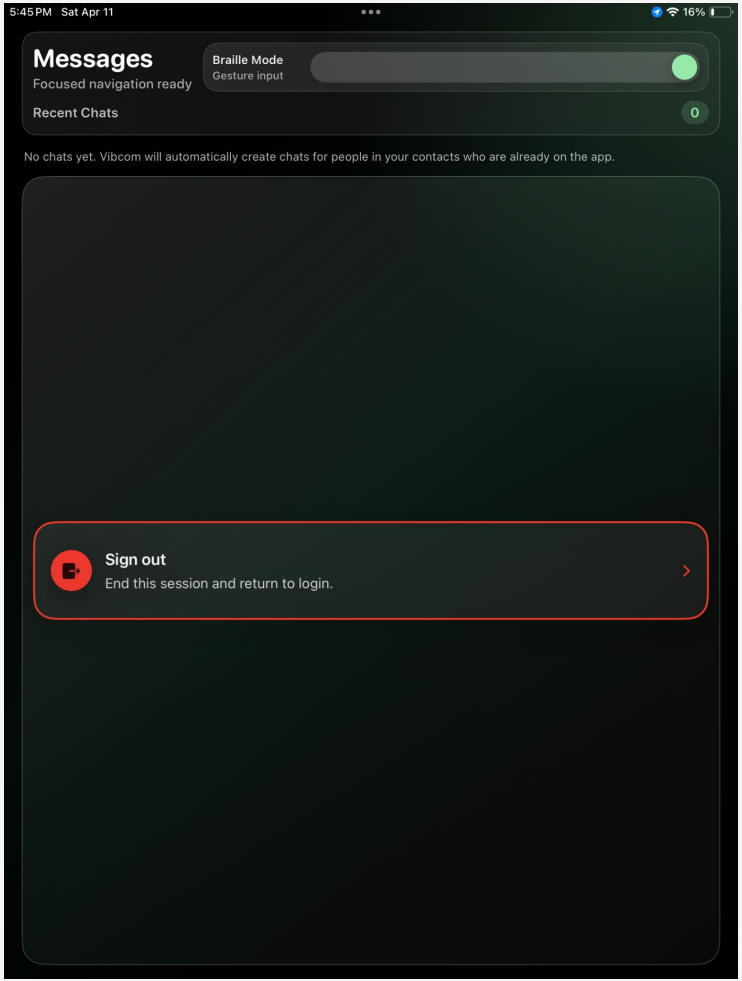
Vibcom Contacts Permissio n



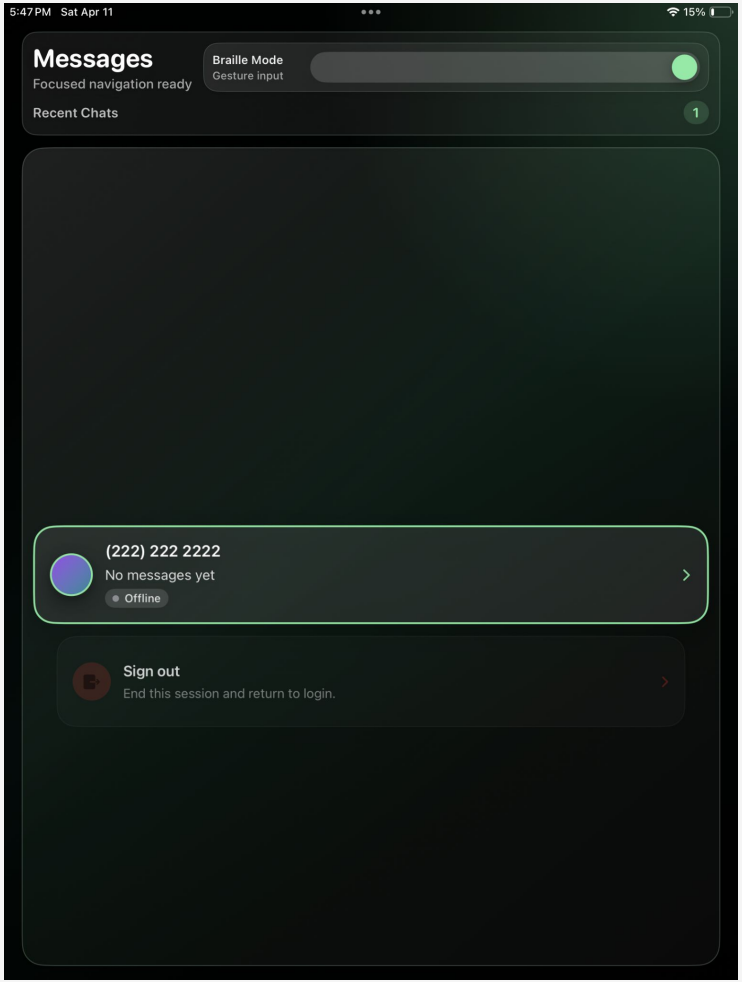
Vibcom Location Permissio n



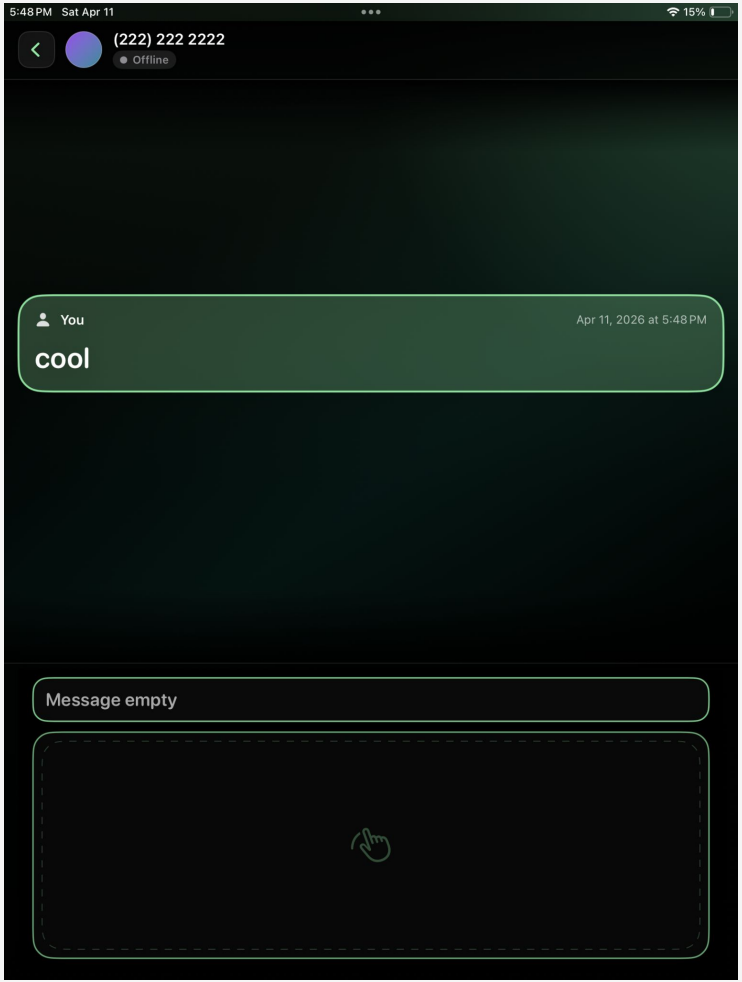
Vibcom Messages Home Screen (Empty)



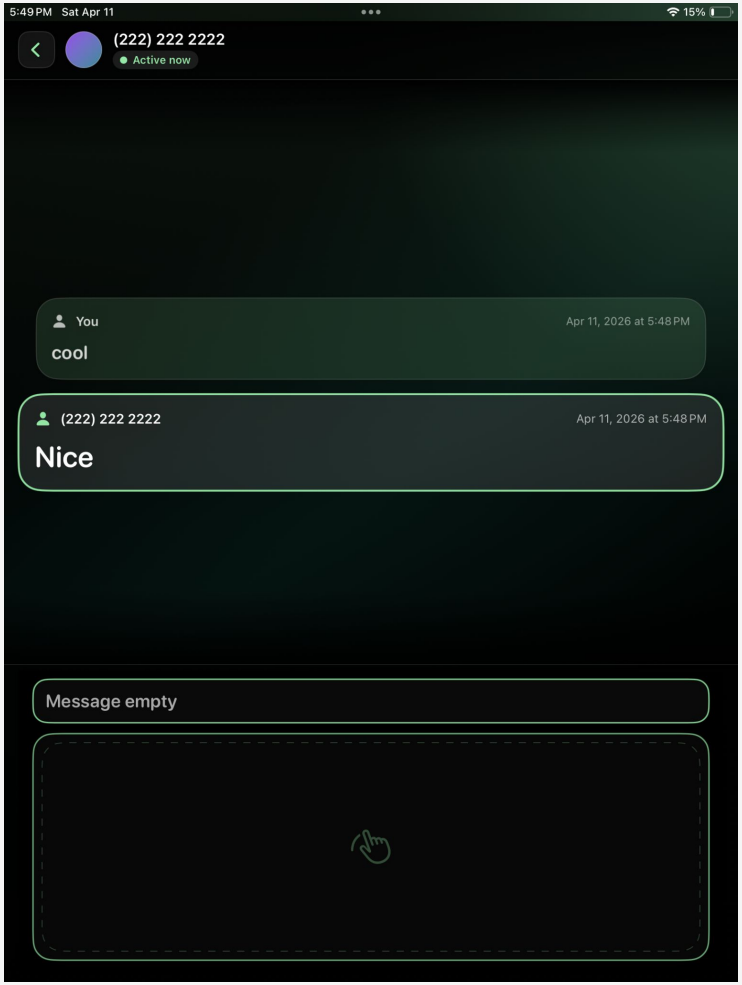
Vibcom Messages Home Screen



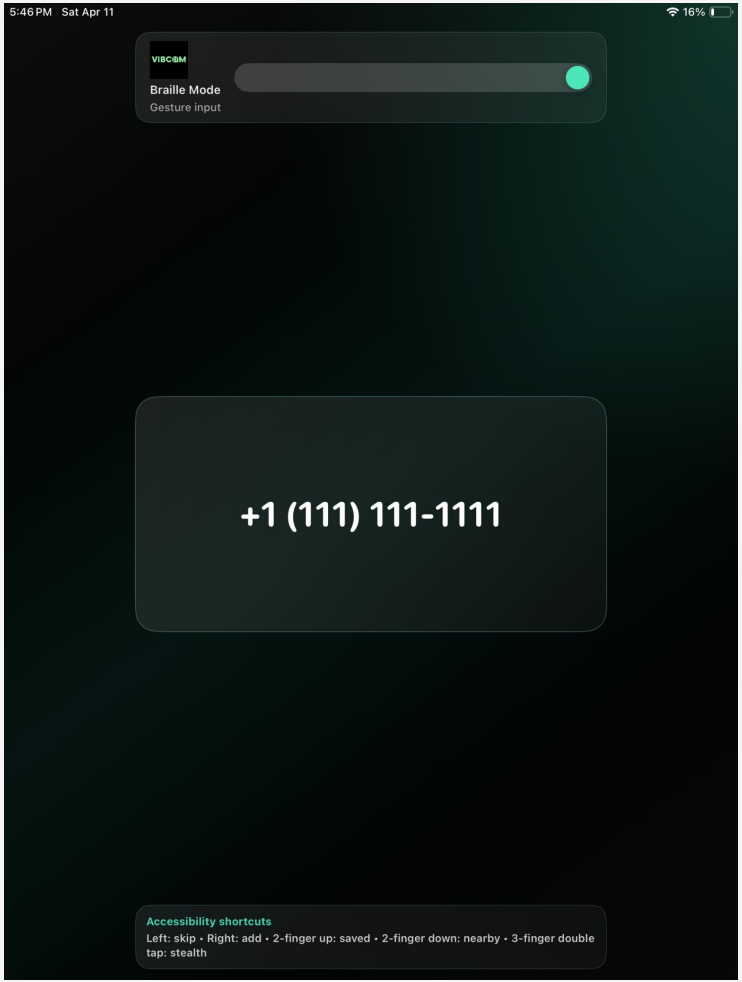
Vibcom Chat Detail (One-Way)



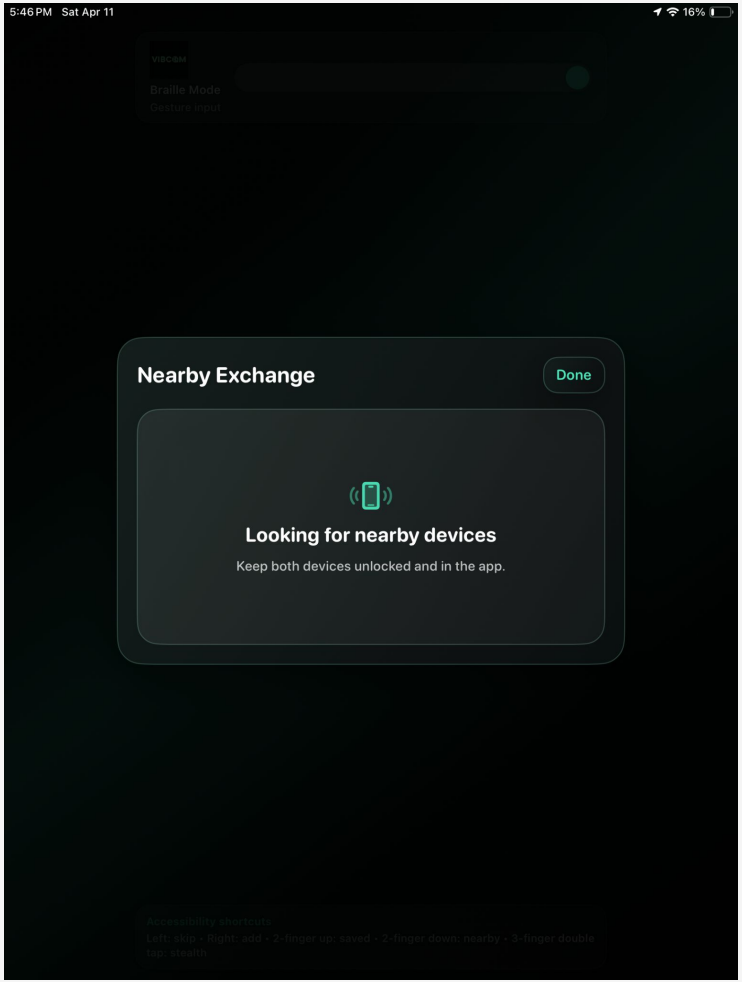
Vibcom Chat Detail (Two-Way)



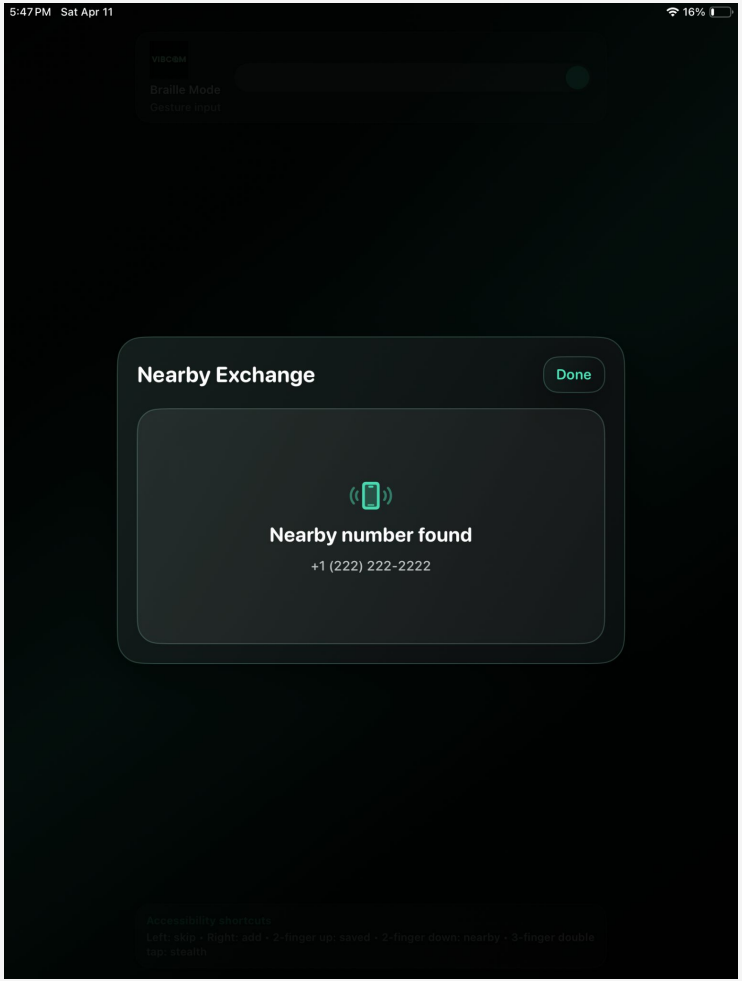
Vibconnect Main Discovery



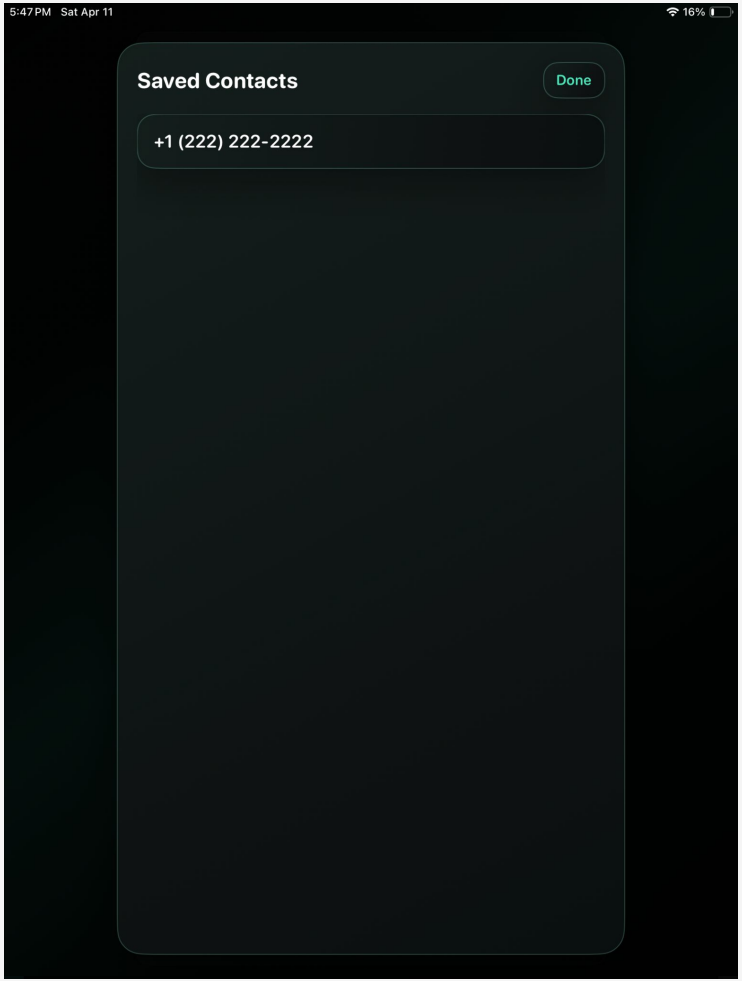
Vibconnect Nearby Exchange (Searching)



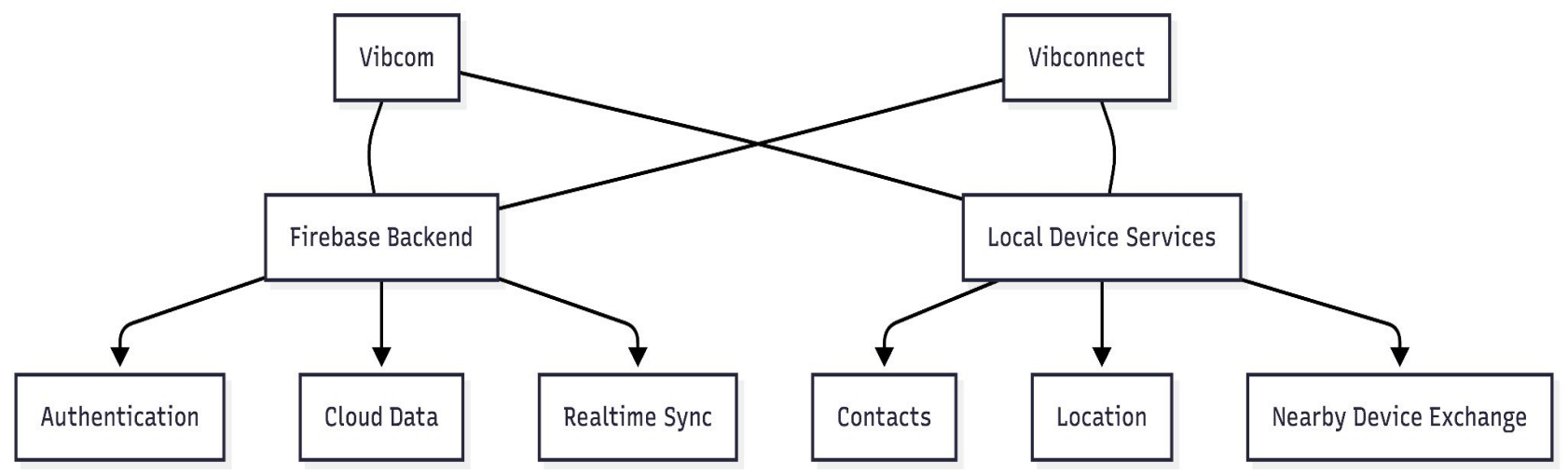
Vibconnect Nearby Exchange (Found)



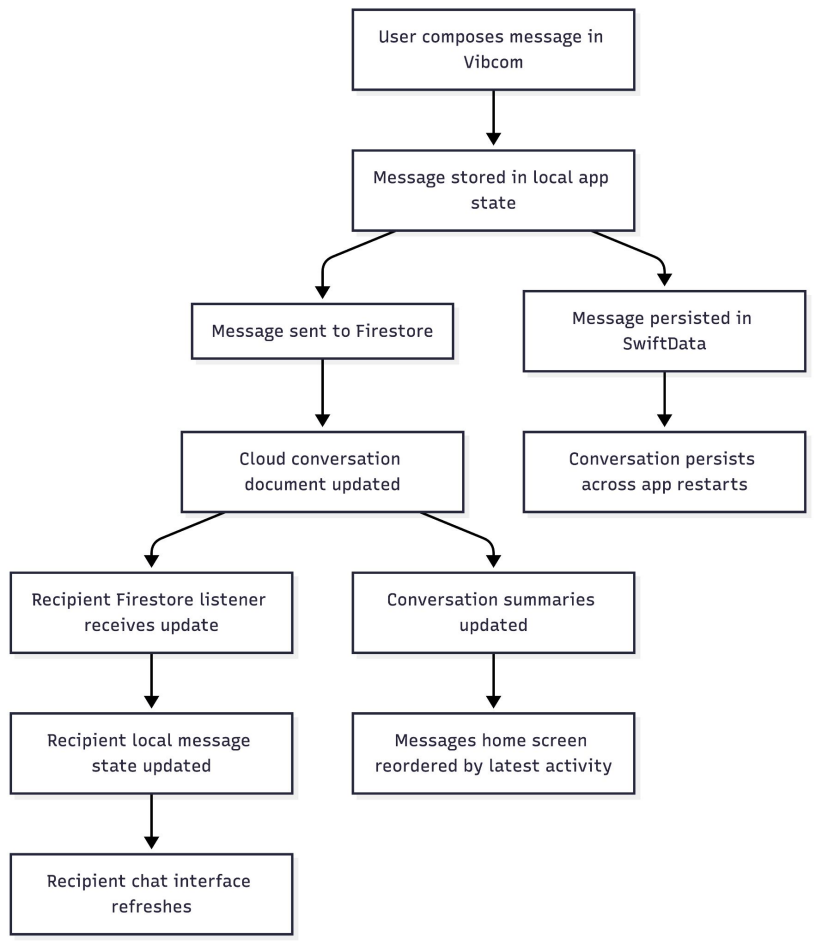
Vibconnect Saved Contacts



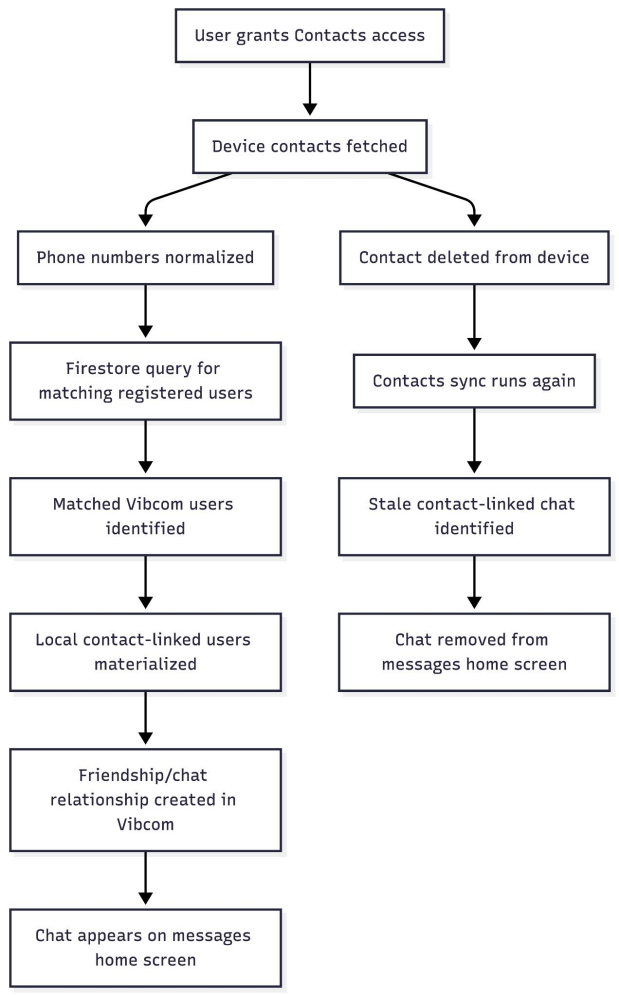
High Level Architecture



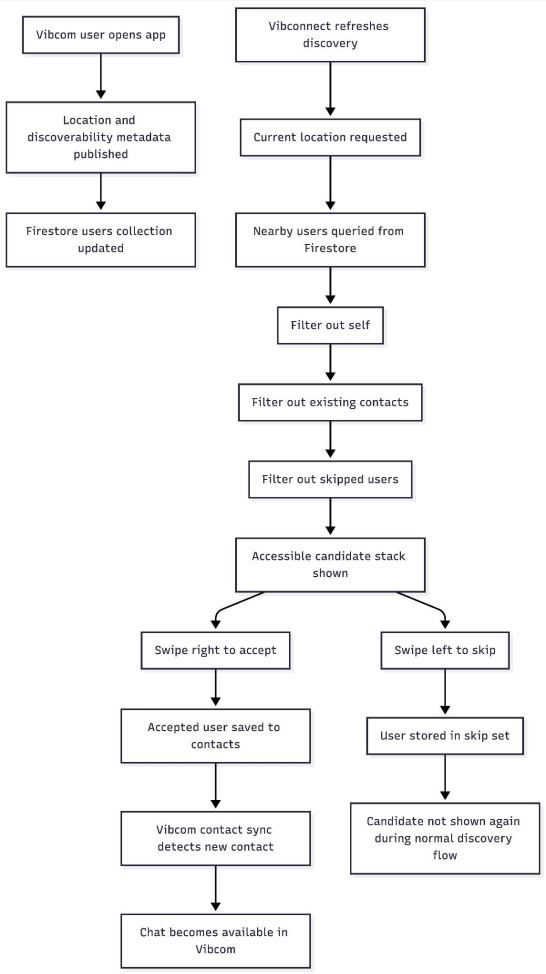
Vibcom Messaging Workflow



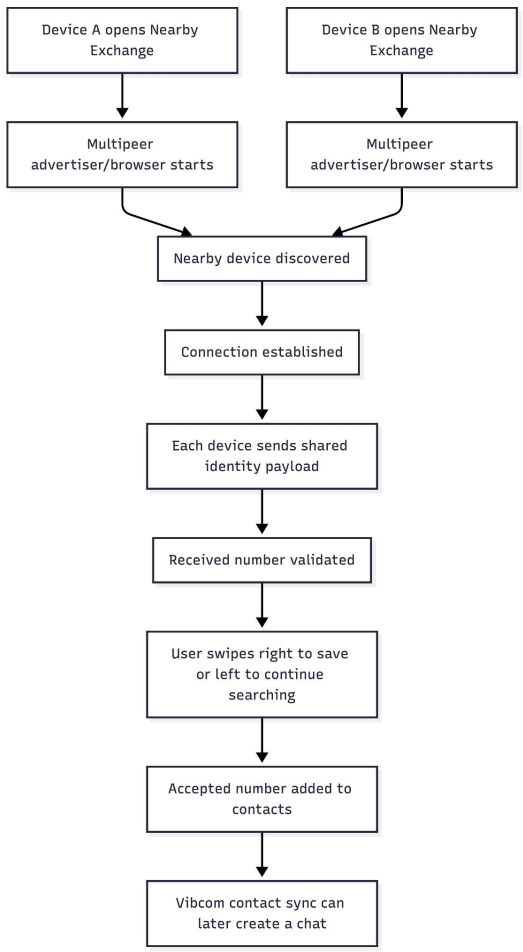
Vibcom Contact Workflow



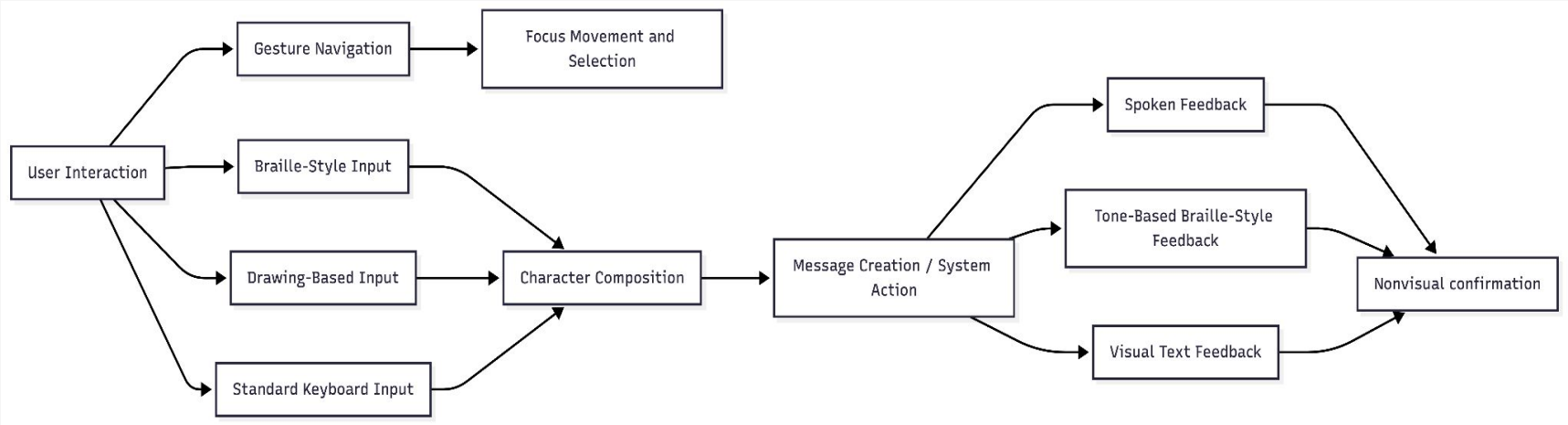
Vibconnect Nearby Discovery Workflow



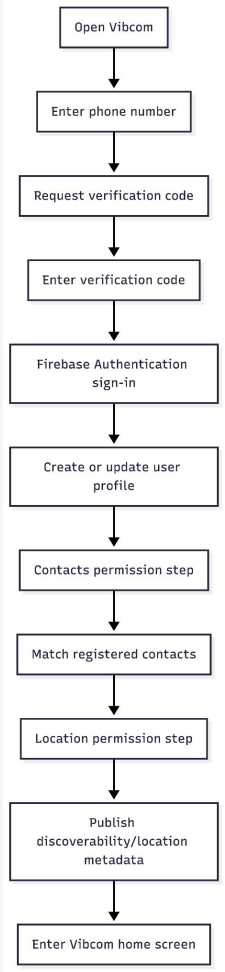
Vibconnect Nearby Exchange Workflow



Accessibility Interaction Model



Vibcom Authentication and Onboarding



Vibcom Real Time Presence

